Data Visualization with Tableau

Answer Booklet for the Question Bank

Section	Marks each	Questions	Question numbers
Section 1	2	66	Q1 – Q66
Section 2	5	33	Q67 – Q99
Section 3	10	15	Q100 – Q114

How to write: 2-mark answers need a definition plus one example or two points. 5-mark answers need a definition, 5–8 points or a comparison table, and an example. 10-mark answers need an introduction, headings with points, a diagram or table, and a conclusion. Draw the diagrams shown here in your answer sheet.

Section 1: Answers to 2-Mark Questions

Part A: Introduction to Data Visualization

Q1. Define data visualization.

Data visualization is the process of presenting data in graphical form such as charts, graphs, maps and dashboards. It helps users understand information quickly, see trends and patterns, and make better decisions.

Q2. List any four reasons why data visualization is important.

- Makes complex data easy to understand.
- Identifies trends and patterns quickly.
- Detects outliers and anomalies.
- Supports faster and better decision-making.

Q3. Name any four common types of charts and state what each is used for.

- **Bar chart**: compare categories.
- **Line chart**: show trends over time.
- **Pie chart**: show proportions/percentages.
- **Scatter plot**: show the relationship between two variables.

Q4. Who was William Playfair? What did he introduce?

William Playfair was an 18th-century pioneer of statistical graphics. He introduced the **bar chart (1786)**, the **line chart** and the **pie chart (1801)**, which became standard tools for comparing data.

Q5. What is a Coxcomb chart? Who used it and why?

A Coxcomb chart (polar area chart) shows values as wedges of a circle. **Florence Nightingale** used it to show the causes of soldiers' deaths in the **Crimean War**, proving that most deaths came from preventable disease.

Q6. How did John Snow use data visualization?

John Snow plotted cholera deaths on a **map of London** and identified the **source of the cholera outbreak** (a water pump). This proved the importance of visualization in epidemiology (study of disease).

Q7. What was René Descartes' contribution to data visualization?

René Descartes introduced the **Cartesian coordinate system** (x-axis and y-axis) in the 17th century. It became the foundation of modern graphs.

Q8. Name any four modern data visualization tools.

Tableau, Microsoft Power BI, Google Looker Studio, Microsoft Excel (also Python libraries like Matplotlib, Seaborn, Plotly and R's ggplot2).

Q9. How do IT and cybersecurity teams use data visualization?

They use it to **track network traffic and system performance** and to **detect security threats and unusual activities**, for example spikes in login failures shown on a dashboard.

Q10. Mention any four professionals who use data visualization.

Business analysts, data scientists, financial analysts and healthcare professionals (also marketing professionals, engineers, government agencies, journalists).

Part B: Getting Data into Tableau

Q11. What is a Tableau data source?

A Tableau data source is the **connection between Tableau and the data** to be analysed. It defines where the data comes from, how it is connected and how Tableau accesses it.

Q12. What is a data connector in Tableau?

A data connector is the mechanism that allows Tableau to connect to a data source and bring its data in for analysis. It acts as a **bridge** between Tableau and external data (e.g. Excel connector, MySQL connector).

Q13. Name the six types of data connectors in Tableau.

1. File-based connectors 2. Database connectors 3. Cloud-based connectors 4. Server connectors 5. Web and online connectors 6. ODBC and JDBC connectors.

Q14. What are ODBC and JDBC?

ODBC (Open Database Connectivity) is a standard way to connect Tableau to databases using an ODBC driver. **JDBC (Java Database Connectivity)** connects to databases using Java-based drivers.

Q15. List any four file types Tableau can connect to.

Excel (.xlsx), CSV (.csv), Text (.txt), JSON (.json) (also PDF, Microsoft Access .mdb/.accdb and spatial files such as Shapefile, KML, GeoJSON).

Q16. What is the Data Interpreter?

The Data Interpreter is an option on the Data Source page that helps Tableau **clean and correctly read** structured data in spreadsheet or text files (e.g. by ignoring extra headers or blank rows).

Q17. What is an RData file? Can Tableau connect to it directly?

An RData file (.RData / .rda) is an R file that stores R objects such as data frames, variables and models. **No**, Tableau cannot connect to it directly; the data is first exported from R to CSV or Excel.

Q18. What is Rserve?

Rserve is a server that lets **Tableau communicate with R**. It is used to run R's statistical calculations or predictive models from Tableau, rather than simply importing an RData file.

Q19. What details are needed to connect Tableau to MySQL?

Server (e.g. localhost), Port (usually 3306), Database name, Username and Password. The MySQL driver must also be installed.

Q20. What is Custom SQL in Tableau?

Custom SQL lets a user **write their own SQL query** to fetch specific data from a database, e.g. SELECT Customer_Name, City, Sales FROM Customer JOIN Sales ON Customer.Customer_ID = Sales.Customer_ID. It is used when the data cannot be obtained by just selecting tables.

Q21. What is a NoSQL database? Give two examples.

A NoSQL (Not Only SQL) database stores large volumes of **unstructured, semi-structured or rapidly changing** data without fixed tables. Examples: **MongoDB, Cassandra** (also DynamoDB, Couchbase, HBase).

Q22. Give two differences between SQL and NoSQL databases.

SQL	NoSQL
Data in structured tables	Data as documents, key-value pairs, graphs or wide columns
Fixed, predefined schema	Flexible schema

Q23. Mention any four security measures used when Tableau connects to a database.

Username and password authentication, database permissions, encryption and row-level security (also integrated authentication and user access controls).

Part C: Live Connection, Extracts and In-Memory

Q24. What is a live connection?

A live connection means Tableau connects **directly** to the original data source and fetches data in real time whenever a chart needs it. The data is not imported; it stays in the database.

Q25. What is an extract in Tableau?

An extract is a **copy (snapshot) of the data** taken from the original source and stored in Tableau's **.hyper** format. It gives faster performance and allows offline analysis, but must be refreshed to show new data.

Q26. What is the Hyper engine?

Hyper is Tableau's **in-memory data engine** used for extracts (.hyper files). It uses columnar storage and is optimized for fast analytical queries on large data.

Q27. What is meant by in-memory processing in Tableau?

In-memory processing means the data is loaded into the computer's **RAM** so Tableau can read, filter and aggregate it very quickly. Tableau does this with Hyper extracts.

Q28. Differentiate full refresh and incremental refresh.

Full refresh	Incremental refresh
The entire extract is replaced with new data	Only new or changed records are added
Takes more time	Faster; less data processed

Q29. What is an extract filter? Give an example.

An extract filter limits which data goes into the extract, making it smaller and faster. Example: the database has sales from 2015–2026 but only **Year = 2024, 2025, 2026** is extracted.

Q30. Give two advantages and two disadvantages of a live connection.

Advantages: near-real-time data; no duplicate copy of data. **Disadvantages:** speed depends on the database and network; the database must always be available.

Part D: Organizing and Combining Data

Q31. What is the data layer in Tableau?

The data layer is the **underlying structure of data** that Tableau uses to connect, combine, prepare and analyse information. It includes data sources, tables, relationships, joins, unions, fields, calculations and filters.

Q32. What is a table? Name its parts.

A table is data arranged in rows and columns. Parts: **rows** (records), **columns** (fields), **field names** (column names) and **data values** (information in cells).

Q33. Differentiate logical tables and physical tables.

Logical tables are connected using relationships on common fields without merging all rows. **Physical tables** are used inside a logical table to create joins and unions.

Q34. What is a union? Give an example.

A union **adds rows** from two or more tables that have the same column structure. Example: January Sales and February Sales tables are combined into one table.

Q35. Define dimensions with examples.

Dimensions are fields containing **qualitative/categorical** data used to describe, group and filter data. Examples: Customer Name, Product, Category, City, Region.

Q36. What is a discrete field? In which colour is it shown?

A discrete field creates **separate, distinct categories** in a view (e.g. Region: North | South | East | West). Discrete fields are shown in **blue**.

Q37. What are facts? Give examples.

Facts are **measurable numerical values** that record business events, usually stored in a fact table. Examples: Sales Amount, Quantity Sold, Profit, Discount, Cost.

Q38. What is a fact table?

A fact table is a table in a data warehouse that stores the **measurements (facts)** such as Sales and Profit, together with keys (Order_ID, Product_ID, Customer_ID) that link to dimension tables.

Q39. What is a join? Name its types.

A join combines data from two or more tables using a **common field**. Types: **Inner, Left, Right and Full Outer** join.

Q40. What is an inner join?

An inner join returns **only the records that match** in both tables. Rows without a match in the other table are left out.

Q41. What is a relationship in Tableau?

A relationship is a **logical connection** between tables based on a common field. The tables stay separate and Tableau decides how to query them, preserving each table's level of detail.

Q42. What is data blending?

Data blending combines data from **two or more different data sources** in one view using a common field (e.g. Region). It is used when sources cannot be joined directly.

Q43. What are primary and secondary data sources in blending?

The **primary** data source is the one used first in the view; it gives the main dimensions and measures. The **secondary** source is the additional source whose data (usually measures) is blended in.

Q44. What is meant by level of detail?

Level of detail is how detailed a view is, decided by the dimensions in it. Sales by Region gives one value per region; adding Product (Region → Product) makes the view more detailed.

Part E: Tableau Files

Q45. What is a Tableau workbook?

A Tableau workbook (.twb or .twbx) is the main Tableau Desktop file used to create, save and share visualizations. It contains data sources, worksheets, dashboards and stories.

Q46. Differentiate .twb and .twbx.

.twb	.twbx
Tableau Workbook	Tableau Packaged Workbook
XML file with workbook structure only; no data	Workbook + data + images + supporting files in one package

Q47. What is a Tableau Packaged Data Source (.tdsx)?

A .tdsx file contains the **data source definition, a local extract of the data and supporting files**. It can be shared and used offline without access to the original database.

Q48. Differentiate .tds and .tdsx.

.tds	.tdsx
Tableau Data Source	Tableau Packaged Data Source
Only the connection definition; no data	Definition + data extract + supporting files

Q49. What is a story in Tableau?

A story is a **sequence of worksheets and dashboards** arranged as story points to present data insights as a narrative. It is created using the Story menu.

Part F: The Tableau Workspace

Q50. What is the Tableau workspace?

The Tableau workspace is the main working environment where users connect to data, create visualizations and build dashboards and stories. It includes the Menu Bar, Toolbar, Data Pane, shelves, Marks Card and Canvas.

Q51. What is the Data Pane?

The Data Pane is on the left side of the workspace. It displays connected data sources, tables, dimensions, measures, sets and parameters, and is used to access and organize fields before building charts.

Q52. What is a calculated field? Give an example.

A calculated field is a **new field created using a Tableau formula**. Example: **Profit Ratio = SUM([Profit]) / SUM([Sales])**.

Q53. What is a hierarchy? Give an example.

A hierarchy groups related dimensions into a **drill-down structure** from higher to lower level. Example: **Country → State → City**.

Q54. What is a parameter?

A parameter is a **user-defined variable** (number, string, date or Boolean) that makes analysis interactive. It can control calculations, filters, reference lines and chart behaviour.

Q55. Differentiate groups and sets.

Group	Set
Combines several values into one category	Custom subset: members are IN or OUT
e.g. Delhi + Punjab = North India	Can be fixed or dynamic

Q56. What is the Analytics Pane?

The Analytics Pane is next to the Data Pane. It provides ready-made analytical objects (constant line, reference line, trend line, forecast, cluster, box plot) that are **dragged onto a chart** without writing formulas.

Q57. Differentiate a constant line and a reference line.

A **constant line** is drawn at a **fixed value** you type (e.g. target = 1,00,000). A **reference line** is **calculated from the data** (average, median, min, max, sum) and changes with the data.

Q58. What is a trend line? Name its types.

A trend line shows the **direction/relationship** in data and helps predict future values. Types: **Linear, Logarithmic, Exponential, Polynomial and Power**.

Q59. What is the use of the Swap button?

Swap **exchanges the fields on the Rows and Columns shelves**. E.g. Rows = Sales, Columns = Region becomes Rows = Region, Columns = Sales, which changes the chart's orientation.

Q60. What is Show Me?

Show Me is a toolbar feature that **recommends suitable chart types** (bar, line, pie, map, scatter, treemap, etc.) based on the fields selected in the Data Pane.

Q61. What is the use of Pause Updates and Run Update?

Pause Updates stops Tableau from redrawing the view after every change (useful with large data). **Run Update** applies all changes and refreshes the view once.

Q62. What is the Canvas in Tableau?

The Canvas (view area) is the central area of a worksheet, dashboard or story where charts, tables, filters and objects are placed. It is where data is turned into visual, interactive information.

Q63. What is the Pages shelf?

The Pages shelf divides a view into a **sequence of pages**, one for each value of a field, with Play, Pause, Previous and Next controls. Example: Year on Pages shows 2019, 2020, 2021... one by one.

Q64. What is a context filter?

A context filter is a filter that is **applied first** and creates a context for other filters; the other filters then work only on the data it keeps. E.g. filter Region first, then Top 10 products within that region.

Q65. What is the Marks Card?

The Marks Card controls how data points (marks) look in a view using **Color, Size, Label, Detail, Shape and Tooltip**. The mark type (bar, line, circle, etc.) is also chosen here.

Q66. Differentiate Keep Only and Exclude.

Keep Only shows only the selected marks and removes the rest (select Technology, see only Technology). **Exclude** removes the selected marks and keeps the rest (select Furniture, Furniture disappears).

Section 2: Answers to 5-Mark Questions

Part A: Introduction to Data Visualization

Q67. Explain the importance and benefits of data visualization with an example.

Data visualization is presenting data as charts, graphs, maps and dashboards so that it is understood quickly.

Importance

- Makes complex data easy to understand.
- Identifies trends and patterns quickly.
- Highlights relationships between variables.
- Detects outliers, anomalies and errors.
- Supports faster, better decision-making.
- Simplifies comparison of datasets.
- Helps monitor business performance through dashboards.
- Enables prediction and forecasting from historical data.

Benefits

- Better understanding of large datasets.
- Faster analysis than reading tables.
- Improved business intelligence and planning.
- Better presentation of research findings.

Example

Month	Jan	Feb	Mar	Apr
Sales	120	150	180	170

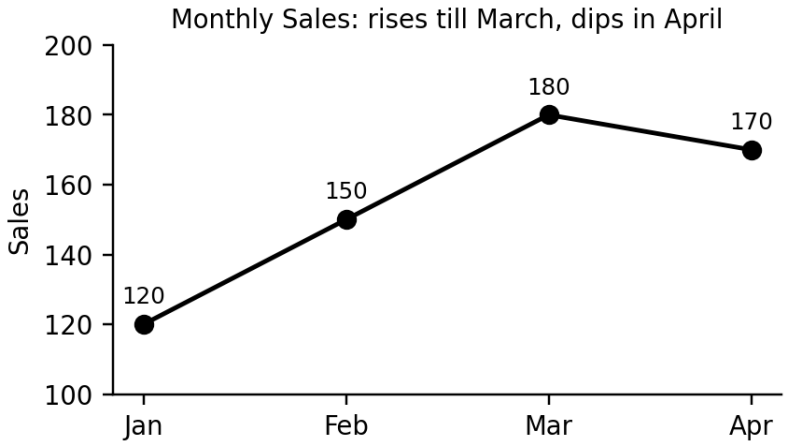


Figure: The same data as a line chart; the trend is visible at a glance

A line chart of this data immediately shows sales rising from January to March and falling slightly in April, which is harder to see in the table.

Q68. Explain the common types of data visualization charts.

Different charts are used for different purposes:

Chart	Purpose	Example
Bar chart	Compare different categories	Sales of each region
Line chart	Show trends over time	Monthly sales
Pie chart	Show proportions or percentages	Market share
Histogram	Show frequency distribution	Number of students in each marks range
Scatter plot	Show relationship between two variables	Advertising vs Sales
Heat map	Show intensity of values with colour	Sales by region and month
Area chart	Show cumulative trends	Total users over time
Box plot	Show spread of data and outliers	Distribution of marks

Choosing the right chart makes the message clear: comparison → bar, trend → line, part-of-whole → pie, relationship → scatter, distribution → histogram/box plot.

Q69. Write a short note on the contributions of Florence Nightingale, John Snow and Charles Minard.

These three 19th-century pioneers showed that visualization can change decisions and save lives.

Florence Nightingale

- Used the **Coxcomb (polar area) chart**.
- Showed the causes of soldiers' deaths in the **Crimean War**.
- Proved most deaths were from preventable disease, which led to better hospital sanitation.

John Snow

- Plotted cholera deaths on a **map of London**.
- The cluster of deaths pointed to the **source of the outbreak** (a water pump).
- Proved the importance of visualization in **epidemiology**.

Charles Joseph Minard

- Created a **flow map of Napoleon's 1812 Russian campaign**.
- Showed army size, route, direction and temperature in one graphic.
- Regarded as one of the **greatest statistical graphics** ever made.

Conclusion: In the 19th century visualization spread into science, economics and public health because of such work.

Q70. Explain who uses data visualization and how (any five users).

Data visualization is used by many professionals to analyse data, communicate insights and support decisions.

1. **Business analysts**: analyse sales, revenue and customer behaviour; create dashboards and reports.
2. **Data scientists**: explore datasets, find patterns, present machine-learning results and predictions.
3. **Financial analysts**: monitor stock prices and financial performance; analyse budgets, profits and investments.

4. **Healthcare professionals**: visualize patient records and disease trends; monitor hospital and public-health data.
5. **Government agencies**: present census, economic and crime statistics; support policy planning.
6. **IT and cybersecurity teams**: track network traffic and system performance; detect security threats and unusual activities.
7. **Marketing professionals**: track campaign performance and market trends.
8. **Media and journalists**: create infographics and explain data to the public.

(Also data analysts, researchers, educators/students and engineers.)

Part B: Getting Data into Tableau

Q71. Explain the steps to get data into Tableau.

Getting data is the first step in Tableau.

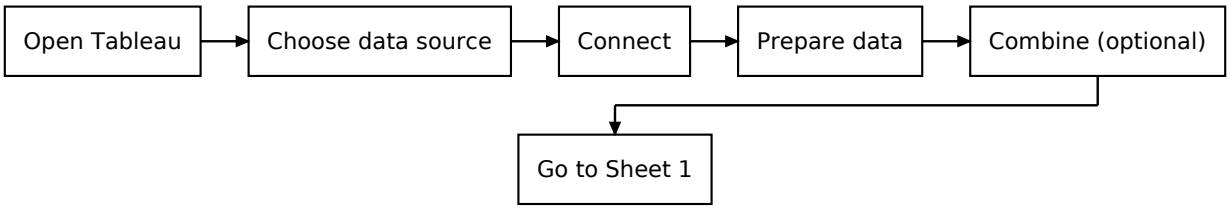


Figure: Steps to get data into Tableau

1. **Open Tableau**: launch Tableau Desktop; the Connect pane appears on the start page.
2. **Choose a data source**: Excel, CSV, PDF, SQL Server, MySQL, Oracle, PostgreSQL, Google Sheets, Microsoft Access or cloud databases (BigQuery, Snowflake, Redshift).
3. **Connect**: select the source, browse and open the file or enter database details; Tableau opens the Data Source page.
4. **Prepare the data**: rename fields, change data types, filter records, hide columns, clean missing or incorrect values.
5. **Combine data (optional)**: joins (common field), relationships (keep tables separate) and unions (append rows).
6. **Go to a worksheet**: click Sheet 1 and drag fields to Rows, Columns, Marks or Filters.

Data can be used through a **live** connection (latest data) or an **extract** (local copy, faster).

Q72. Explain the types of data sources Tableau can connect to.

A Tableau data source is the connection between Tableau and the data. Tableau can connect to:

Type	Examples
Spreadsheet files	Microsoft Excel, CSV, text files
Databases	MySQL, SQL Server, Oracle, PostgreSQL
Cloud databases	Google BigQuery, Amazon Redshift, Snowflake
Cloud applications	Salesforce, Google Analytics
Web data	Web Data Connector, JSON, PDF
Statistical files	SAS, SPSS, R data
Big Data sources	Hadoop, Apache Hive, Spark

Each source can be used as a **live** connection (direct, real-time) or an **extract** (local snapshot, faster, offline). Many sources can be used in one workbook, e.g. sales from Excel and customers

from MySQL related on Customer ID.

Q73. Explain the types of data connectors in Tableau.

A **data connector** is the bridge that allows Tableau to connect to a data source.

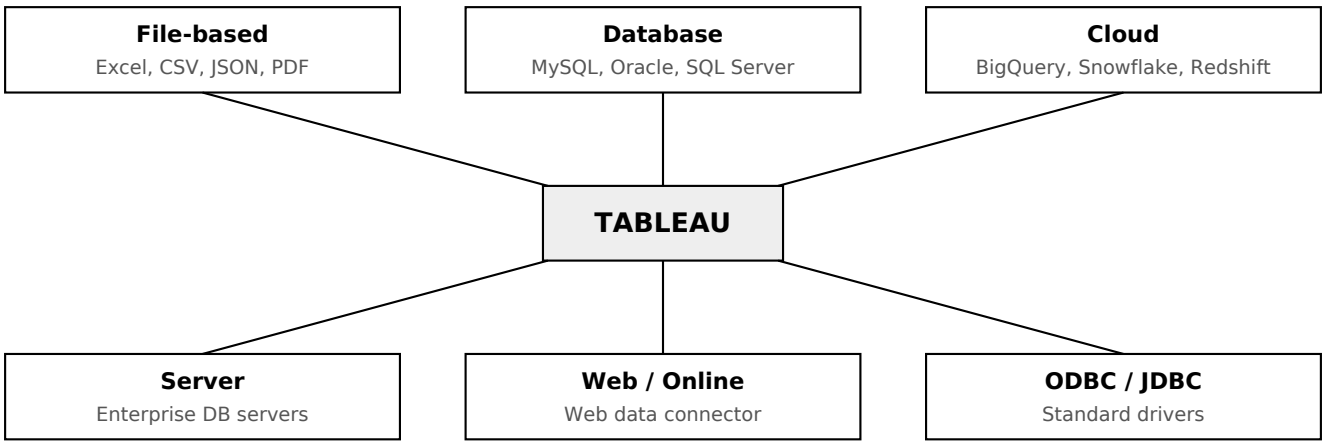


Figure: Types of data connectors (the connector is the bridge to Tableau)

1. **File-based connectors**: files on a computer/network; Excel, CSV/Text, JSON, PDF, spatial files.
2. **Database connectors**: relational databases; MySQL, SQL Server, Oracle, PostgreSQL, IBM DB2, SQLite.
3. **Cloud-based connectors**: cloud platforms; Google BigQuery, Amazon Redshift, Snowflake, Azure SQL, Salesforce.
4. **Server connectors**: enterprise database servers holding centralized, large-scale data.
5. **Web and online connectors**: online services and web platforms (Web Data Connector).
6. **ODBC and JDBC connectors**: standard drivers; ODBC = Open Database Connectivity, JDBC = Java Database Connectivity.

Working: Data Source → Connector → Tableau → Data Preparation → Visualization.

Q74. Explain the steps to connect a text (CSV) file to Tableau. Mention its advantages and limitations.

Tableau can connect directly to CSV, TXT and TSV files.

Steps

1. Open Tableau Desktop.
2. Start page → **Connect → To a File → Text File**.
3. Browse to the .csv / .txt / .tsv file, select it and click **Open**.
4. The Data Source page opens; check field names, data types, delimiters, rows and columns.
5. Change data types or rename fields if required (Data Interpreter can help clean data).
6. Go to a worksheet and build charts (e.g. Sales by Region from Sales.csv).

Advantages

- Simple; no database server needed.
- Good for small/medium data; easy to share; good for learning.

Limitations

- Large files may be slow.
- Data must be properly structured.
- File changes need a refresh; no advanced database features.

Q75. How can data stored in an RData file be used in Tableau? Explain.

An **RData file (.RData / .rda)** stores R objects such as data frames and models. Tableau **cannot connect to it directly**, so the data is exported first.

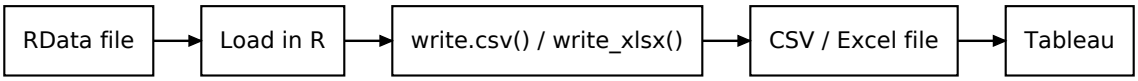


Figure: Getting RData into Tableau

Method 1: Export to CSV

```
load("sales_data.RData")
write.csv(sales_data, "sales_data.csv", row.names = FALSE)
```

Tableau: Connect → To a File → Text File → sales_data.csv

Method 2: Export to Excel

```
library(writexl)
write_xlsx(sales_data, "sales_data.xlsx")
```

Tableau: Connect → To a File → Microsoft Excel → sales_data.xlsx

Method 3: Rserve

Tableau can communicate with R through **Rserve** to run R's statistical or predictive calculations from Tableau.

Q76. Explain the steps to connect Tableau with a MySQL database.

MySQL is a relational database; Tableau connects to it directly.

1. Open Tableau Desktop; under **Connect** select **MySQL**.
2. Install the MySQL driver if asked.
3. Enter details: **Server** (e.g. localhost), **Port** (usually 3306), **Database**, **Username**, **Password**.
4. Click **Sign In**.
5. Tableau shows databases and tables; drag the required table (e.g. Sales from SalesDB) to the Data Source area.
6. Add more tables and create relationships, joins or unions if needed.
7. Choose Live or Extract.
8. Click Sheet 1 and build charts (Sales by Region, Profit by Product).

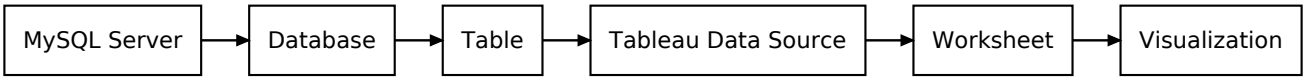


Figure: MySQL connection flow (e.g. localhost:3306 → SalesDB → Sales)

Q77. Differentiate a database and a Tableau data source.

A **database** stores an organization's data; a **Tableau data source** defines how Tableau uses that data.

Database	Tableau data source
Stores organizational data	Defines how Tableau accesses and uses data
Managed by a DBMS	Managed through Tableau
Designed for storage and transactions	Designed for analysis and visualization
Can contain many tables	Contains selected tables and relationships

Database	Tableau data source
Examples: MySQL, Oracle, SQL Server	Examples: .tds, .tdsx, or a connection inside a workbook

Example: a MySQL database may have 50 tables; the Tableau data source may use only Customer and Sales tables related on Customer_ID.

Q78. Explain NoSQL databases and how Tableau connects to them.

NoSQL (Not Only SQL) databases store large volumes of unstructured, semi-structured or rapidly changing data with a flexible schema.

Database	Data model
MongoDB	Document
Cassandra	Wide-column
Amazon DynamoDB	Key-value / document
Couchbase	Document / key-value
HBase	Wide-column

Connecting

NoSQL Database → Connector/Driver → Tableau → Data Source → Worksheet → Dashboard. Steps: open Connect, choose the connector, enter server and login details, select the collection, handle nested fields, build charts.

Why use NoSQL with Tableau

Handles huge data, semi/unstructured data and changing structures; useful for Big Data; can be combined with other sources.

SQL vs NoSQL: SQL uses structured tables with a fixed schema; NoSQL uses documents, key-value, graph or wide-column structures with a flexible schema.

Part C: Live Connection, Extracts and In-Memory

Q79. Explain live connection with its advantages and disadvantages.

A **live connection** connects Tableau directly to the original data source and fetches data in real time; the data is not imported.

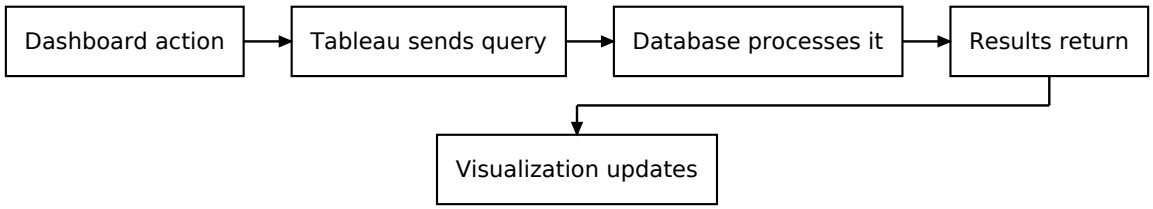


Figure: How a live connection works

Features

Real-time data, data stays in the database, queries processed by the database server, automatic updates, needs database availability.

Advantages

- Near-real-time data.
- No duplicate storage of data.

- Suitable for frequently changing data.
- Handles very large databases.
- Data management stays centralized.

Disadvantages

- Performance depends on the database and network.
- Database must be available.
- Complex dashboards send many queries and increase database load.
- Offline analysis is not possible.

Example: a live sales dashboard showing the latest orders and inventory.

Q80. Differentiate live connection and extract.

A **live connection** queries the original data directly; an **extract** is a copy of the data stored as a .hyper file.

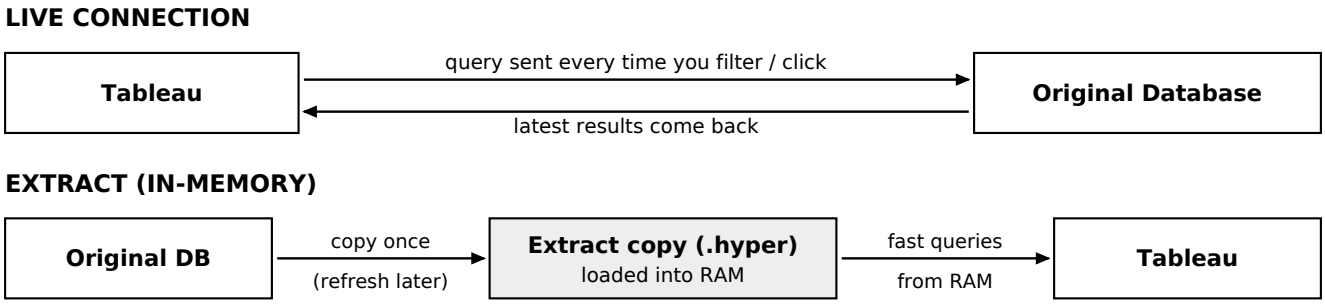


Figure: Live connection vs Extract

Feature	Live connection	Extract
Data location	Original database	Tableau extract (.hyper)
Data freshness	Near real-time	Current only after refresh
Performance	Depends on database/network	Often faster
Database load	Higher	Lower during analysis
Offline access	Limited	Better
Refresh needed	No	Yes (full or incremental)
Best for	Frequently changing data	Fast interactive analysis of large data

Q81. Explain in-memory processing in Tableau with its advantages and limitations.

In-memory means storing data in **RAM** so Tableau can access it very fast. Tableau uses the **Hyper engine** with extracts.

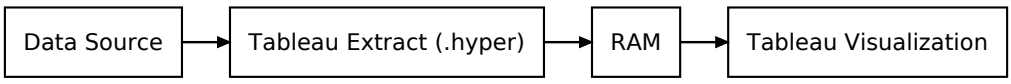


Figure: How in-memory works

Features

Fast processing, high-performance analytics (calculations, filters, aggregation, sorting), columnar storage, less database dependency, supports large datasets, extract refresh.

Advantages

- Faster dashboards.
- Efficient filtering and aggregation.
- Fewer repeated queries to the source.
- Helps when the source database is slow or far away.

Limitations

- Not real-time unless refreshed.
- Needs storage space and a lot of memory for big extracts.
- Source changes are not shown until refresh.

Example: 10 million SQL records → Hyper extract → fast dashboard.

Q82. Explain data extraction in Tableau and the steps to create an extract.

Data extraction is creating a copy of data from the original source in Tableau's optimized **.hyper** format.

Why extract?

Better performance, faster filtering and aggregation, works when the source is unavailable, fewer database queries, data can be filtered before extracting, supports scheduled refresh.

Steps

1. Connect to a data source (Excel, CSV, MySQL).
2. Go to the Data Source page.
3. Select **Extract** instead of Live.
4. Configure extract settings and optionally add extract filters (e.g. Year 2024–2026).
5. Tableau creates a .hyper file (e.g. Sales_Data.hyper).
6. Open a worksheet and build charts.

Refresh types

Full refresh replaces the whole extract; **incremental refresh** adds only new/changed records.

Part D: Organizing and Combining Data

Q83. Explain the components of the data layer in Tableau.

The **data layer** is the foundation of data that Tableau uses to connect, combine, prepare and analyse information.

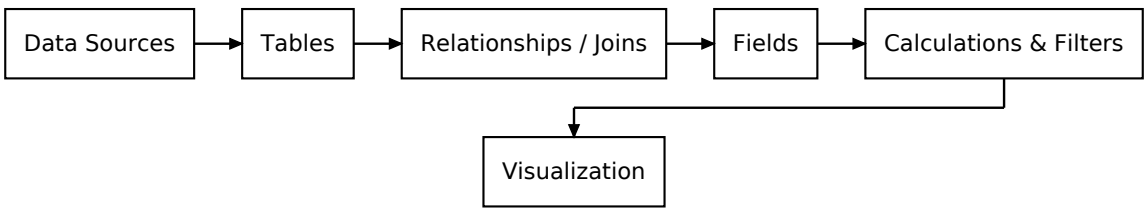


Figure: Data layer flow

1. **Data sources**: Excel, CSV, MySQL, SQL Server, Oracle, Google Sheets.
2. **Tables**: data in rows and columns.
3. **Relationships**: connect tables on common fields (Customer ID).
4. **Joins**: combine tables on a common field (inner, left, right, full outer).
5. **Unions**: append rows of similar tables (Jan + Feb sales).

6. **Dimensions and measures**: categories vs numbers.
7. **Data types**: text, number, date, Boolean.
8. **Calculated fields**: e.g. Profit Ratio = SUM([Profit]) / SUM([Sales]).
9. **Filters**: e.g. only 2026 or one region.

Q84. Explain the types of tables in Tableau and the ways to combine them.

A **table** is data in rows and columns (rows = records, columns = fields).

Types of tables

- **Database tables**: in MySQL, Oracle, SQL Server, PostgreSQL.
- **Spreadsheet tables**: each Excel worksheet is a table.
- **Logical tables**: connected by relationships without merging rows.
- **Physical tables**: used inside logical tables for joins and unions.

Ways to combine tables

- **Relationships**: connect on related fields, keep original structure.
- **Joins**: add columns using a common field.
- **Unions**: add rows from tables with similar columns.

JOIN = adds COLUMNS (tables side by side, matched on ID)

Customer

ID	Name
101	Rahul
102	Priya

Sales

ID	Sales
101	50,000
102	30,000

→

Result (wider)

ID	Name	Sales
101	Rahul	50,000
102	Priya	30,000

UNION = adds ROWS (similar tables one below the other)

Jan Sales

Month	Sales
Jan	120

Feb Sales

Month	Sales
Feb	150

→

Result (longer)

Month	Sales
Jan	120
Feb	150

Figure: Join vs Union

Q85. Explain dimensions and their purpose in Tableau with an example.

Dimensions are qualitative/categorical fields that describe and classify data (Customer Name, Product, Category, City, Region). They are usually **discrete** and shown in **blue**.

Purpose

Categorize data, group records, filter data, create headers, define the level of detail, compare categories, organize the view.

Example

Product	Region	Sales
Laptop	North	50,000
Laptop	South	60,000
Printer	North	25,000

Product and Region are dimensions, Sales is a measure. Region on Rows + Sales on Columns shows sales for each region.

Level of detail

Sales by Region gives one value per region; adding Product (Region → Product) makes the view more detailed. Adding dimensions increases detail.

Q86. Explain facts with a fact table example. How are facts related to measures?

Facts are measurable numerical values that record business events, stored in the **fact table** of a data warehouse.

Order_ID	Product_ID	Customer_ID	Quantity	Sales	Profit
1001	P01	C101	2	50,000	8,000
1002	P02	C102	5	20,000	4,000

Order_ID, Product_ID and Customer_ID are keys; Quantity, Sales and Profit are facts.

Facts and measures

When Tableau connects, numeric fact fields become **measures** and are aggregated: Sales → SUM(Sales), Quantity → SUM(Quantity), Discount → AVG(Discount). Functions: SUM, AVG, MIN, MAX, COUNT, MEDIAN.

Fact table vs dimension table

The fact table stores measurements; dimension tables (e.g. Product: P01 Laptop Technology) describe them.

Importance: facts give the numbers for KPIs, charts and trends (Region + Sales → Sales by Region).

Q87. Differentiate facts and dimensions.

Facts are measurable numbers; **dimensions** are descriptive information about them.

Facts	Dimensions
Represent measurable values	Represent descriptive information
Usually numerical	Usually categorical
Used for calculations and aggregation	Used for grouping and filtering
e.g. Sales, Profit, Quantity	e.g. Product, Region, Customer
Stored in fact tables	Stored in dimension tables
Usually become Tableau measures (green)	Usually become Tableau dimensions (blue)

Example: in "Sales by Region", Sales is the fact and Region is the dimension.

Q88. Differentiate joins and relationships in Tableau.

A **join** physically combines tables; a **relationship** logically connects them.

Join	Relationship
Combines tables physically at the data-source level	Defines a logical connection between tables
Uses a specific join type (inner, left, right, full outer)	Tableau decides how to query related tables
Can create duplicate rows in many-to-many cases	Preserves each table's level of detail better
Useful for detailed row-level combining	Preferred in Tableau's logical data model
Works on physical tables	Works on logical tables

Example: Customer and Sales tables linked on Customer_ID can be either joined (one combined table) or related (two tables, linked).

Q89. Differentiate data blending and joins.

Data blending combines different data sources in one view; a **join** combines tables usually within the same source.

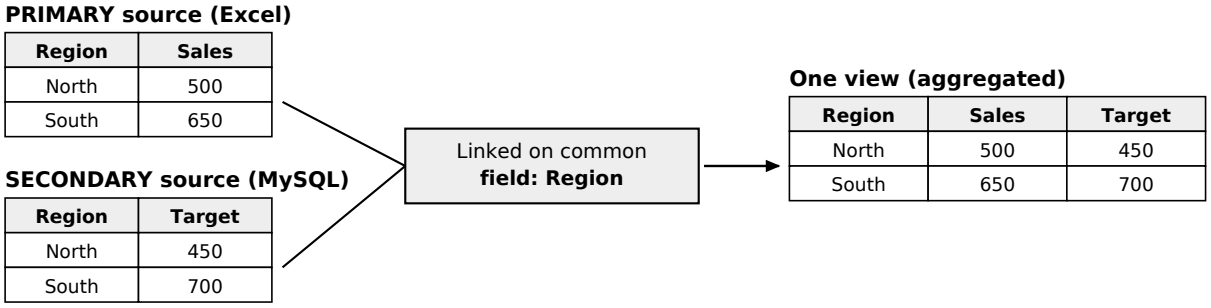


Figure: Data blending, with a primary and a secondary source shown together

Data blending	Joins
Combines data from different data sources	Usually combines tables within one data source
Uses primary and secondary sources	Uses tables with join conditions
Combines at the visualization/aggregated level	Combines at the row level
Useful when sources cannot be joined directly	Useful when tables have compatible structures
Uses common linking fields	Uses join keys

Part E: Tableau Files

Q90. Explain the components of a Tableau workbook and its file types.

A **Tableau workbook** is the main Tableau Desktop file for creating, saving and sharing visualizations.

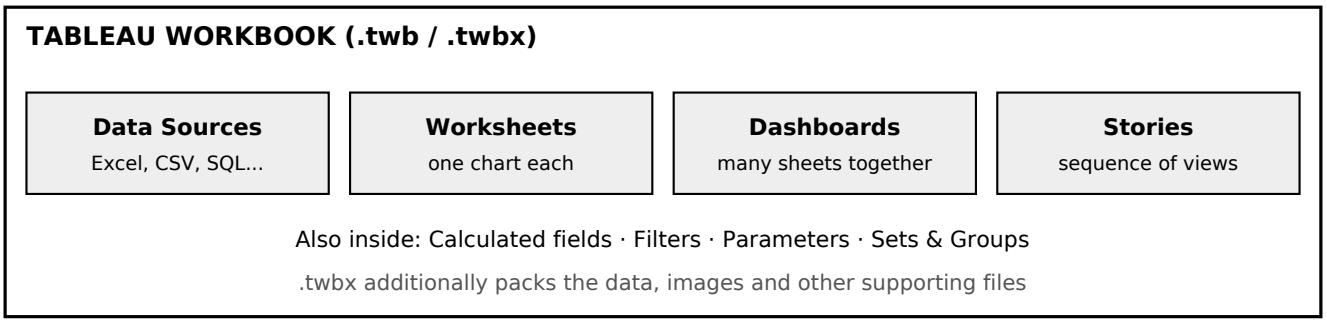


Figure: What a Tableau workbook contains

Components

1. Data sources: Excel, CSV, SQL, cloud connections.
2. Worksheets: individual charts or tables.
3. Dashboards: many worksheets in one interactive view.
4. Stories: a sequence of views telling a data story.
5. Calculated fields: custom formulas.
6. Filters: restrict data.
7. Parameters: user-controlled variables.

8. Sets and groups: organize data.

File types

Extension	Description
.twb	Tableau Workbook, XML structure only, no data
.twbx	Packaged Workbook: workbook + data + images + files, easy to share

Q91. Explain the Tableau Packaged Data Source (.tdsx) with its components, advantages and limitations.

A **.tdsx** file contains the data source definition, a local extract of the data and supporting files, so it can be shared without the original database. (A .tds file has only the definition, no data.)

Components

Connection information, data extract, calculated fields, groups and sets, hierarchies, metadata (field names, aliases, data types), supporting files.

Advantages

- Easy to share with colleagues.
- No access to the original database needed.
- Offline analysis.
- Everyone uses the same version of data.

Limitations

- Data becomes outdated unless the extract is refreshed.
- File size can be large.
- Source changes are not reflected automatically.

Example: an analyst saves an Excel sales extract as .tdsx; the team uses it without the Excel file.

Part F: The Tableau Workspace

Q92. Explain any five menus of Tableau Desktop.

The Menu Bar has 11 menus: File, Data, Worksheet, Dashboard, Story, Analysis, Map, Format, Server, Window, Help.

1. **File**: manage workbooks; New, Open, Save, Save As, Close, Export, Print.
2. **Data**: manage data sources; Connect to Data, New Data Source, Refresh, Edit Relationships, Extracts, Replace data source.
3. **Worksheet**: manage sheets; New Worksheet, Duplicate, Clear, Show/Hide Cards, Show Summary, Export.
4. **Dashboard**: New Dashboard, Add Sheets, Actions, Layout, Export.
5. **Analysis**: Aggregate Measures, % of Total, Totals, Trend Lines, Forecast, Reference Lines, Calculated Field, Parameter.
6. **Format**: Font, Alignment, Borders, Shading, Lines, Cell Size.
7. **Server**: Sign In, Publish Workbook, Publish Data Source, Permissions.

(Story, Map, Window and Help can also be written.)

Q93. Explain the important toolbar buttons in Tableau.

The toolbar is below the Menu Bar and gives one-click access to common commands.

- **New / Open / Save**: workbook operations; save as .twb or .twbx.
- **Undo / Redo**: reverse / re-apply the last action.
- **Pause Updates / Run Update**: stop automatic redraw with large data, then update once.
- **Swap**: exchange Rows and Columns fields.
- **Sort Ascending / Descending**: A–Z/low–high or Z–A/high–low.
- **Group**: combine marks (Bangalore + Chennai + Hyderabad = South India).
- **Keep Only / Exclude**: show only / remove selected marks.
- **Highlight**: emphasize selected data.
- **Show Me**: recommends chart types.
- **Presentation Mode**: clean full-screen view for presenting.

Menu Bar vs Toolbar: the menu bar has all commands in text menus; the toolbar has frequently used commands as icons.

Q94. Differentiate the Data Pane and the Analytics Pane.

Feature	Data Pane	Analytics Pane
Purpose	Organizes and manages data fields	Adds analytical features to visualizations
Contents	Dimensions, Measures, Parameters, Sets, Groups	Trend lines, Forecasts, Reference lines, Box plots, Clusters
Usage	To build visualizations	To enhance and analyse visualizations
Main function	Data preparation and field management	Statistical and predictive analysis
Interaction	Drag fields to Rows, Columns, Marks, Filters	Drag analytical objects onto the view
Location	Data tab on the left	Analytics tab beside the Data tab

Example: Sales and Region are dragged from the Data Pane to make a bar chart; an Average Line is then dragged from the Analytics Pane onto it.

Q95. Explain the features of the Canvas in Tableau.

The **Canvas** is the central area of a worksheet, dashboard or story where visualizations are built and arranged.

1. **Visualization area**: displays the chart.
2. **Drag-and-drop**: fields dropped from the Data Pane.
3. **Marks**: data shown as bars, circles, lines, shapes.
4. **Formatting**: colours, fonts, labels, borders.
5. **Interactive elements**: filters, parameters, tooltips, actions.
6. **Dashboard arrangement**: position and resize sheets, filters, text, images.
7. **Multiple views**: combine worksheets in a dashboard.
8. **Sorting and filtering**.
9. **Annotations**: highlight important points.
10. **Responsive design**: layouts for different screen sizes.

In a worksheet it holds one chart; in a dashboard, many sheets; in a story, a sequence of views; in a map view, geographic data.

Q96. Differentiate the Columns shelf and the Rows shelf with an example.

The Columns and Rows shelves define the layout of a chart.

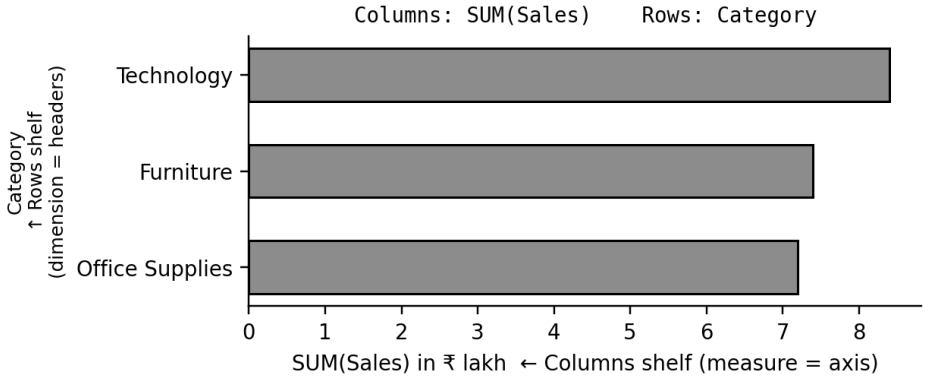


Figure: Sales on Columns + Category on Rows = horizontal bar chart

Columns shelf	Rows shelf
Controls the horizontal arrangement	Controls the vertical arrangement
Creates columns and horizontal axes	Creates rows and vertical axes
Fields displayed left to right	Fields displayed top to bottom
Compares categories horizontally	Organizes categories vertically

A dimension on a shelf creates headers; a measure creates an axis.

Example: Category on Rows and Sales on Columns gives a horizontal bar chart of Sales by Category; Order Date on Columns and Sales on Rows gives a sales trend line.

Q97. Explain the Filters shelf and the types of filters in Tableau.

The **Filters shelf** restricts the data shown so only data meeting conditions appears. Filters can use values, ranges, dates or conditions, apply to one or many worksheets, and can be shown as an interactive filter card.

Example: Country on Filters, select India → only Indian sales are shown.

Types of filters

1. **Dimension filter**: categorical data, e.g. Region = East.
2. **Measure filter**: numeric range, e.g. Sales above ₹10,000.
3. **Date filter**: dates/periods, e.g. year 2024.
4. **Top N filter**: top or bottom N records, e.g. Top 10 products.
5. **Context filter**: applied first; other filters work within it.
6. **Condition filter**: based on a condition, e.g. Profit greater than 0.

Q98. Differentiate the Filters shelf and the Pages shelf.

The **Filters shelf** decides **what data you see**; the **Pages shelf** lets you see the data **one page at a time**.

Filters shelf	Pages shelf
Restricts the data displayed	Divides the view into sequential pages
Removes/hides data not meeting criteria	Shows different values one page at a time
Used for focused analysis	Used for sequential or time-based analysis
Displays a filter card control	Provides Play, Pause, Previous, Next controls

Filters shelf	Pages shelf
e.g. show only India	e.g. show 2019, 2020, 2021... one by one

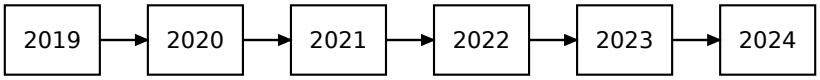


Figure: Pages shelf with Year: press Play to flip through years

Q99. Explain the features of the Analytics Pane.

The **Analytics Pane** (next to the Data Pane) gives drag-and-drop analytical objects.

1. **Constant line**: fixed value, e.g. target ₹1,00,000, pass mark 40.
2. **Reference line**: average, median, min, max, sum or custom value.
3. **Distribution band**: shaded range showing spread.
4. **Box plot**: min, Q1, median, Q3, max and outliers.
5. **Trend line**: linear, logarithmic, exponential, polynomial, power.
6. **Forecast**: predicts future values (sales, demand).
7. **Cluster**: groups similar points using machine learning.
8. **Totals**: subtotals and grand totals.
9. **Average line**: compares values with the mean.
10. **Median with quartiles**: median and quartile ranges.

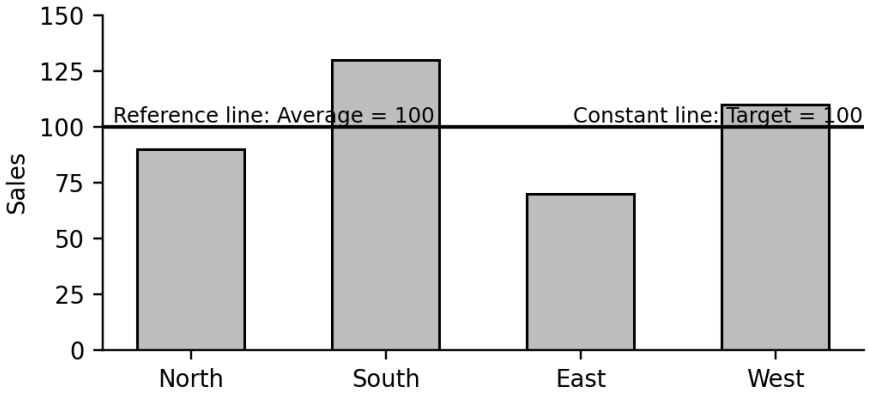


Figure: Constant line (fixed target) vs reference line (average of the data)

Section 3: Answers to 10-Mark Questions

Part A: Introduction to Data Visualization

Q100. What is data visualization? Explain why data is visualized, its benefits, common types and applications.

Introduction

Data visualization is the process of presenting data in graphical formats such as charts, graphs, maps and dashboards. It turns raw numbers into pictures so that users understand information quickly and make better decisions.

Why visualize data? (importance)

1. Makes complex data easy to understand.
2. Identifies trends and patterns quickly.
3. Highlights relationships between variables.
4. Detects outliers and anomalies.
5. Supports faster and better decision-making.
6. Simplifies comparison of different datasets.
7. Improves communication of information.
8. Helps monitor business performance using dashboards.
9. Makes reports attractive and interactive.
10. Enables prediction and forecasting from historical data.

Benefits

- Better understanding of large datasets.
- Faster analysis than reading tables.
- Easy identification of errors and inconsistencies.
- Improved business intelligence and strategic planning.
- Enhanced presentation of research findings.

Common types of visualization

Chart	Use
Bar chart	Compare categories
Line chart	Show trends over time
Pie chart	Show proportions/percentages
Histogram	Show frequency distribution
Scatter plot	Show relationship between two variables
Heat map	Show intensity of values
Area chart	Show cumulative trends
Box plot	Show spread and outliers

Applications

Business analytics, healthcare, finance, education, scientific research, marketing, government statistics and weather forecasting.

Example

Month	Jan	Feb	Mar	Apr
Sales	120	150	180	170

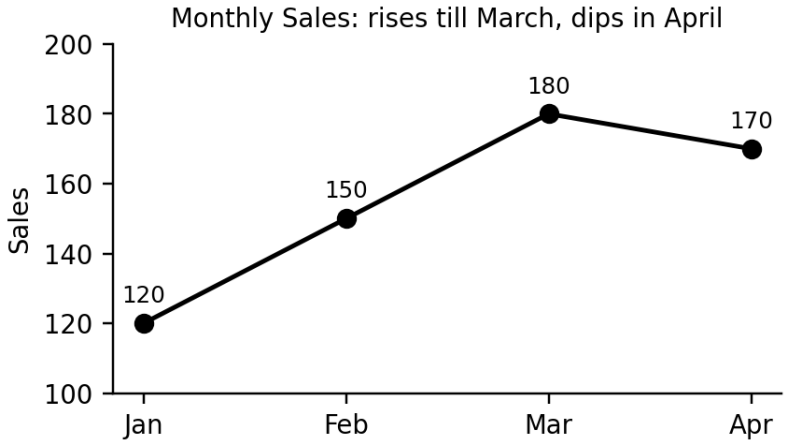


Figure: The same data as a line chart; the trend is visible at a glance

The line chart shows at a glance that sales grew from January to March and fell slightly in April.

Conclusion

Data visualization converts data into insight. It saves time, improves understanding and supports evidence-based decisions in every field.

Q101. Explain the history of data visualization from ancient times to the modern era.

Introduction

Data visualization is the practice of representing information graphically. Its history spans thousands of years, from simple maps to today's interactive dashboards.

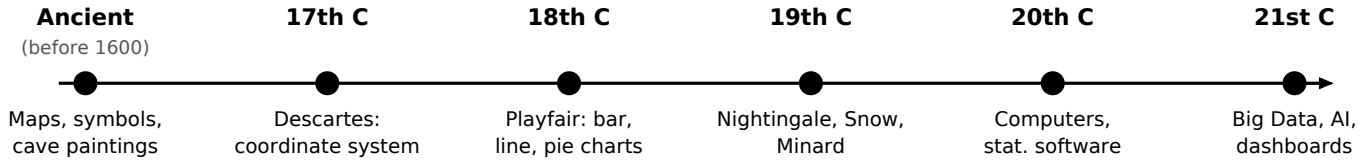


Figure: Timeline of data visualization

1. Ancient period (before 1600)

- Early humans used cave paintings and symbols to record information.
- Egyptians and Babylonians created maps for land measurement and navigation.
- Greek scholars used geometric diagrams to explain mathematics.

2. 17th century

- Scientific discoveries increased the need for visual representation.
- Astronomers and mathematicians used graphs and coordinate systems.
- **René Descartes** introduced the **Cartesian coordinate system**, the foundation of modern graphs.

3. 18th century

- Modern statistical graphics emerged.

- **William Playfair** introduced the **bar chart (1786)**, **line chart** and **pie chart (1801)**, which became standard tools.

4. 19th century

- Visualization expanded into science, economics and public health.
- **Florence Nightingale**: Coxcomb (polar area) chart on causes of death in the Crimean War.
- **John Snow**: map locating the source of the London cholera outbreak, a landmark for epidemiology.
- **Charles Joseph Minard**: flow map of Napoleon's 1812 Russian campaign, one of the greatest statistical graphics.

5. 20th century

- Computers transformed visualization; statistical software created charts automatically.
- Businesses adopted dashboards and reporting systems; interactive visualization became possible.

6. 21st century

- Big Data, AI and Machine Learning increased demand for advanced visuals.
- Interactive dashboards allow real-time analysis.
- Tools: Tableau, Power BI, Google Looker Studio, Excel, Python (Matplotlib, Seaborn, Plotly), R (ggplot2).

Summary table

Period	Major development
Ancient	Maps, symbols, diagrams
17th century	Cartesian coordinate system
18th century	Playfair's bar, line and pie charts
19th century	Nightingale, Snow, Minard
20th century	Computer charts, statistical software
21st century	Interactive dashboards, Big Data, AI analytics

Importance / Conclusion

This development improved understanding of complex data, enabled evidence-based decisions, advanced science and medicine, supported business intelligence and made information accessible to everyone.

Part B: Getting Data into Tableau

Q102. Explain the complete process of getting data into Tableau, including types of data connections and connection modes.

Introduction

Getting data is the **first step** in Tableau. Tableau connects to many sources, then lets the user clean and prepare the data for visualization.

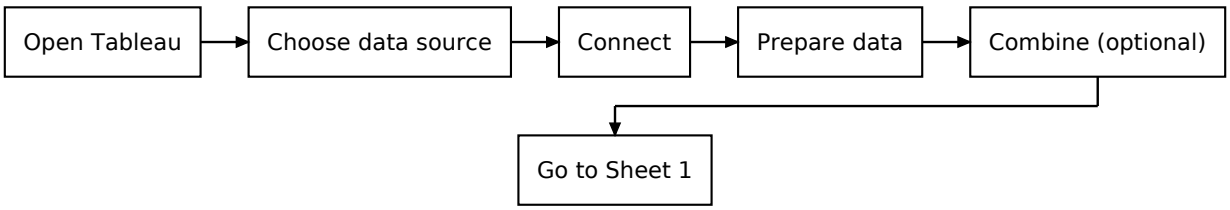


Figure: Steps to get data into Tableau

Steps

1. **Open Tableau**: launch Tableau Desktop; the Connect pane appears.
2. **Choose a data source**: Excel, CSV, PDF, SQL Server, MySQL, Oracle, PostgreSQL, Google Sheets, Microsoft Access, cloud databases.
3. **Connect**: open the file or enter server details; Tableau shows the Data Source page.
4. **Prepare the data**: rename fields, change data types, filter unnecessary records, hide columns, clean missing/incorrect values (Data Interpreter helps).
5. **Combine data (optional)**: joins, relationships, unions.
6. **Go to a worksheet**: click Sheet 1; drag fields to Rows, Columns, Marks or Filters.

Types of data connections

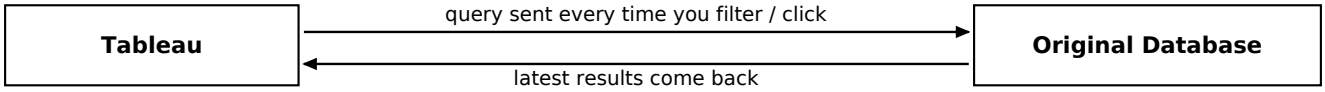
Type	Examples
File-based	Excel, CSV, JSON, PDF
Database	MySQL, SQL Server, Oracle, PostgreSQL
Cloud	Google Sheets, BigQuery, Snowflake, Amazon Redshift

Ways to combine data

- **Join**: combine tables using a common field (adds columns).
- **Relationship**: connect tables but keep them separate.
- **Union**: append rows from tables with the same structure.

Connection modes

LIVE CONNECTION



EXTRACT (IN-MEMORY)



Figure: Live connection vs Extract

- **Live connection**: connects directly to the original source, always latest data, best for frequently updated data; depends on database speed.
- **Extract connection**: creates a local copy (.hyper), faster for large data, works offline, refreshed when needed.

Advantages

Connects to many sources, supports live and extract, drag-and-drop interface, handles large data efficiently, built-in data preparation.

Conclusion

Correctly connecting and preparing data is the base on which all Tableau charts and dashboards are built.

Q103. Explain data connectors in Tableau: their types, how they work and the connection modes.

Introduction

Data connectors are the mechanisms that let Tableau connect to different data sources and bring data in. They act as a **bridge** between Tableau and external data.

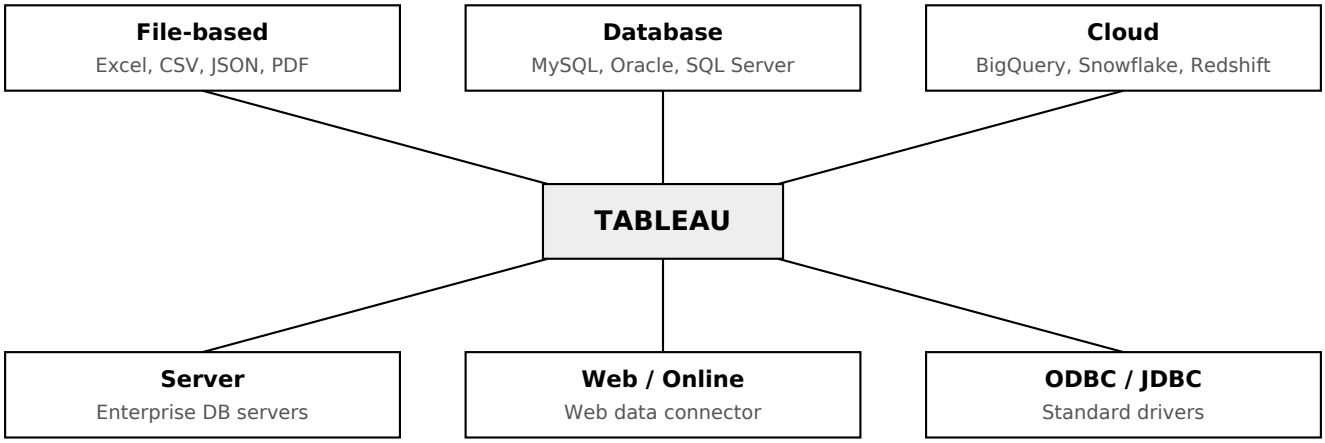


Figure: Types of data connectors (the connector is the bridge to Tableau)

Types of data connectors

1. **File-based connectors**: files on a computer/network (Excel, CSV/Text, JSON, PDF, spatial files).
2. **Database connectors**: relational databases (MySQL, SQL Server, Oracle, PostgreSQL, IBM DB2, SQLite).
3. **Cloud-based connectors**: cloud platforms (BigQuery, Redshift, Snowflake, Azure SQL, Salesforce).
4. **Server connectors**: enterprise database servers with centralized, large-scale data.
5. **Web and online connectors**: online services and web platforms (Web Data Connector).
6. **ODBC and JDBC connectors**: ODBC gives a standard driver-based connection; JDBC uses Java-based drivers.

Common connectors

Connector	Type	Use
Excel	File	Spreadsheet analysis
CSV	File	Text/tabular data
MySQL	Database	Relational analysis
SQL Server	Database	Enterprise data
Oracle	Database	Large organizational DBs
Salesforce	Cloud	CRM data
Snowflake / Redshift	Cloud	Cloud data warehouse
Google BigQuery	Cloud	Large-scale analytics

How a connector works

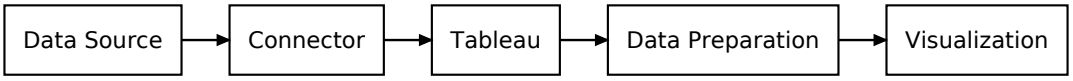


Figure: How a data connector works

Example: Excel File → Excel Connector → Tableau → Data Cleaning → Charts/Dashboard.

Connection modes

- **Live**: Tableau queries the original source whenever data is needed; latest data.
- **Extract**: Tableau stores a copy in its extract format; faster, offline access.

Conclusion

Connectors make Tableau flexible: the same tool can analyse a small CSV file or a huge cloud data warehouse.

Q104. Explain databases in Tableau: types, connecting, live vs extract, working with tables, custom SQL and security.

Introduction

A **database** is a structured collection of data stored in a DBMS. Tableau connects to databases to build charts and dashboards directly from organizational data.

Types of databases supported

MySQL, Microsoft SQL Server, Oracle, PostgreSQL (relational); Amazon Redshift, Snowflake, Google BigQuery, Azure SQL (cloud); Teradata (data warehouse); SAP HANA (in-memory); MongoDB (NoSQL via connectors).

Connecting a database

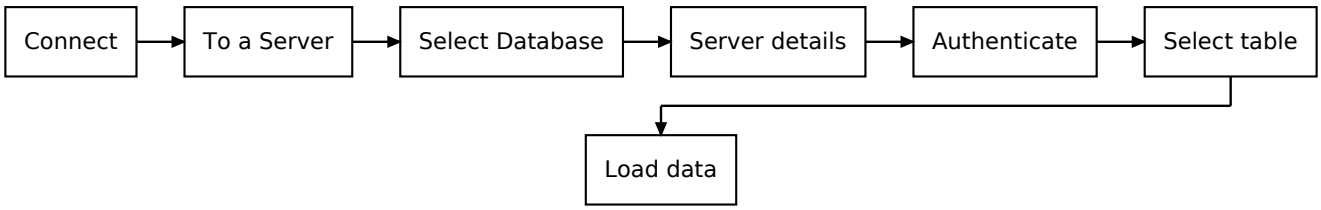


Figure: Connecting a database to Tableau

Details required: server name/IP, port number, database name, username, password, authentication method.

Live vs Extract

Mode	Advantages	Disadvantages
Live	Up-to-date data; no full copy; suits changing data	Depends on database/network; complex queries slow
Extract	Faster; less dependency on DB; supports offline	Not current until refreshed; needs storage

Working with tables

Select tables, create relationships, joins and unions, write custom SQL, apply filters, create calculated fields. Example: Customer (Customer_ID, Name, City) and Sales (Customer_ID, Product, Sales) related on Customer_ID allow sales by customer or city.

Custom SQL

```
SELECT Customer_Name, City, Sales
FROM Customer JOIN Sales
ON Customer.Customer_ID = Sales.Customer_ID;
```

Used when needed data cannot be obtained by selecting tables.

Security

Username/password, integrated authentication, database permissions, encryption, row-level security, user access controls.

Database vs Tableau data source

Database	Tableau data source
Stores data; managed by DBMS	Defines how Tableau uses data; managed in Tableau
For storage and transactions	For analysis and visualization
e.g. MySQL, Oracle	e.g. .tds, .tdsx

Conclusion

Databases let Tableau analyse large organizational data directly, combine many tables and build interactive dashboards without copying data into spreadsheets.

Part C: Live Connection, Extracts and In-Memory

Q105. Compare live connection, extract and in-memory processing in Tableau in detail.

Introduction

After connecting to a source, Tableau can either query the data **live** or create an **extract** that is processed **in memory** by the Hyper engine.

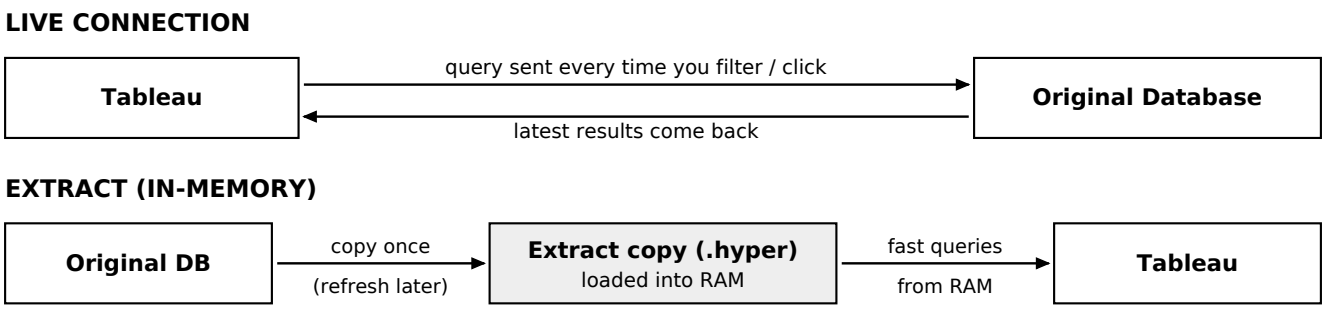


Figure: Live connection vs Extract

1. Live connection

- Tableau connects directly to the original source; data is not imported.
- Every filter/click sends a query to the database, which returns results.
- Real-time data, automatic updates, no duplicate storage.
- Needs the database to be available; speed depends on database and network; increases database load.

2. Extract

- A copy (snapshot) of data saved as a **.hyper** file.

- Created on the Data Source page by choosing Extract; can apply extract filters (e.g. Year 2024–2026).
- Refreshed by **full** refresh (replace everything) or **incremental** refresh (add new rows).
- Faster, fewer database queries, offline analysis; not current until refreshed; needs storage.

3. In-memory

- Extract data is loaded into **RAM** and processed by the **Hyper engine**.
- Columnar storage, fast calculations, filtering, aggregation and sorting.
- Ideal for interactive dashboards, especially when the source is slow or far away.
- Large extracts need memory and processing resources.

Comparison table

Feature	Live connection	Extract / In-memory
Data location	Original database	.hyper extract (loaded into RAM)
Freshness	Near real-time	As of last refresh
Query processing	Database server	Tableau Hyper engine
Speed	Depends on database/network	Often faster
Database load	Higher	Lower
Offline use	Limited	Supported
Refresh	Not needed	Full or incremental
Best for	Frequently changing data	Fast analysis of large data

Example

10 million sales records: a live dashboard queries SQL every time; an extract copies the data once to Sales_Data.hyper and answers queries from memory.

Conclusion

Choose **live** when data must be current; choose **extract/in-memory** when speed and offline access matter.

Part D: Organizing and Combining Data

Q106. What is a join? Explain the types of joins in Tableau with examples and a diagram.

Introduction

A **join** combines data from two or more tables based on a **common field**. It is used when related information is stored in different tables.

Example tables (common field Customer_ID)

Customer_ID	Customer_Name	City
101	Rahul	Delhi
102	Priya	Mumbai
103	Amit	Bangalore

Customer_ID	Product	Sales
101	Laptop	50000
102	Mobile	30000

Customer_ID	Product	Sales
104	Tablet	20000

Customer = left table, Sales = right table.

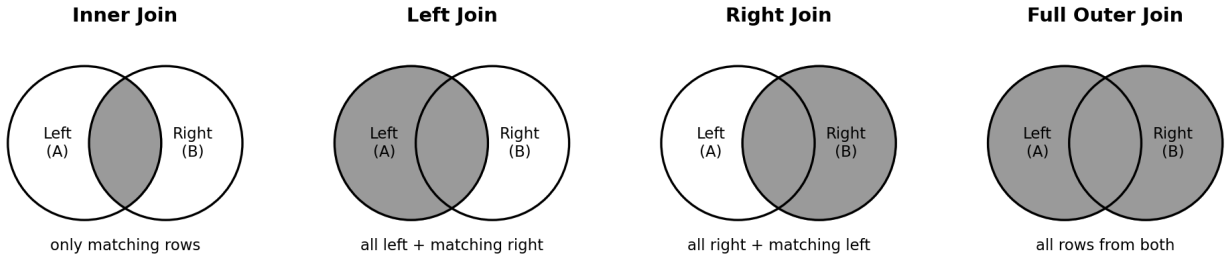


Figure: The four join types (shaded part = rows kept in the result)

1. Inner join

Returns only matching records: 101 Rahul Laptop 50000; 102 Priya Mobile 30000. (103 and 104 excluded.) Use when only matching data is needed.

2. Left join

All records of the left table + matching right: 101, 102 and **103 Amit NULL NULL**. Use to keep every record of the main table.

3. Right join

All records of the right table + matching left: 101, 102 and **104 NULL Tablet 20000**. Use to keep every record of the right table.

4. Full outer join

All records from both tables, NULL where no match:

Customer_ID	Name	Product	Sales
101	Rahul	Laptop	50000
102	Priya	Mobile	30000
103	Amit	NULL	NULL
104	NULL	Tablet	20000

Steps to create a join

Connect to data → Data Source page → drag first table → drag second table next to it → choose join type → select the common field → add conditions if needed → build the view.

Join vs Relationship

Join	Relationship
Physical combination at data-source level	Logical connection
Specific join type	Tableau decides how to query
May duplicate rows (many-to-many)	Preserves level of detail

Conclusion

Joins let us bring related data together; choosing the right type decides which rows are kept.

Q107. Explain data blending in detail. How is it different from joins?

Introduction

Data blending combines data from **two or more different data sources** in a single visualization. It is useful when the sources cannot be joined directly, e.g. sales in Excel and targets in MySQL, blended on **Region**.

How it works

- Uses a **primary data source** and one or more **secondary data sources**.
- **Primary**: the source used first in the view; provides the main dimensions and measures.
- **Secondary**: additional source that contributes extra measures.
- Tableau links them on a **common field** and combines the **aggregated** results.

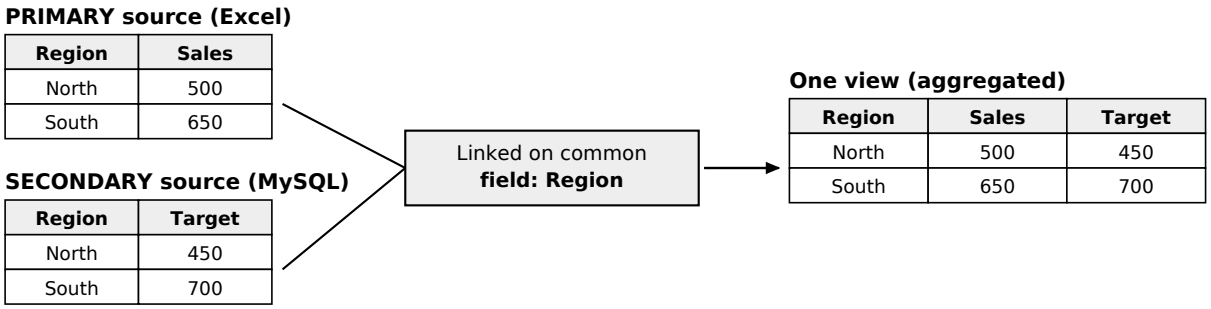


Figure: Data blending, with a primary and a secondary source shown together

Steps

1. Connect to the first data source.
2. Connect to the second data source.
3. Create a worksheet using the primary source.
4. Add a field from the secondary source.
5. Tableau identifies the common field.
6. The common field establishes the blending relationship.
7. Tableau shows aggregated results from both sources.

Example

Primary: Region → Sales. Secondary: Region → Target. Result: Region → Sales → Target, so actual sales can be compared with targets.

Advantages

- Combines data from different sources and systems.
- Useful when direct joins are not possible.
- Works with databases, Excel files and others.
- Does not physically merge the underlying data.

Data blending vs Joins

Data blending	Joins
Different data sources	Tables within one data source
Primary and secondary sources	Tables with join conditions
Combined at visualization / aggregated level	Combined at row level
Used when sources cannot be joined	Used when tables have compatible structures
Common linking fields	Join keys

Conclusion

Blending combines summarized data from different sources through a common field (Region, Product, Customer ID or Date), without merging the data physically.

Q108. Explain facts and dimensions with a fact table and dimension table (star schema) example.

Introduction

In a data warehouse, data is organized into **fact tables** (numbers) and **dimension tables** (descriptions). Tableau treats facts mostly as **measures** and descriptive fields as **dimensions**.

Facts

- Measurable values recording business events: Sales Amount, Quantity, Profit, Discount, Cost.
- Numeric; aggregated using SUM, AVG, MIN, MAX, COUNT, MEDIAN (Sales → SUM(Sales), Discount → AVG(Discount)).

Dimensions

- Qualitative/categorical fields that describe data: Product, Customer, Region, City, Category, Date.
- Used to group, filter, create headers and set the level of detail; usually discrete (blue).

Star schema example

DIMENSION: Product

Product_ID	Name	Category
P01	Laptop	Tech
P02	Printer	Tech

DIMENSION: Customer

Cust_ID	Name	City
C101	Ravi	Delhi
C102	Anil	Mumbai

FACT TABLE: Sales (the numbers)

Order_ID	Product_ID	Cust_ID	Sales	Profit
1001	P01	C101	50,000	8,000
1002	P02	C102	20,000	4,000

DIMENSION: Date

Order_ID	Date	Year
1001	05-Jan	2026

DIMENSION: Region

Cust_ID	Region	State
C101	North	Delhi

Figure: Star schema. The fact table in the centre holds the numbers (facts); the dimension tables around it hold descriptions and link to it through ID keys.

The fact table (Order_ID, Product_ID, Cust_ID, Sales, Profit) sits in the centre; Product, Customer, Date and Region dimension tables around it describe the IDs.

Facts vs Dimensions

Facts	Dimensions
Measurable values	Descriptive information
Numerical	Categorical
Calculation and aggregation	Grouping and filtering
Sales, Profit, Quantity	Product, Region, Customer
Fact tables	Dimension tables
Tableau measures (green)	Tableau dimensions (blue)

Using them together in Tableau

Region + Sales → Sales by Region; Product + Profit → Profit by Product; Year + Sales → Sales trend. Adding more dimensions (Region → Product) increases the level of detail.

Conclusion

Facts provide the numbers and dimensions give them meaning; every chart combines at least one of each.

Q109. Explain the data layer and tables in Tableau, including relationships, joins and unions.

Introduction

The **data layer** is the underlying structure Tableau uses to connect, combine, prepare and analyse data. **Tables** are its basic building blocks.

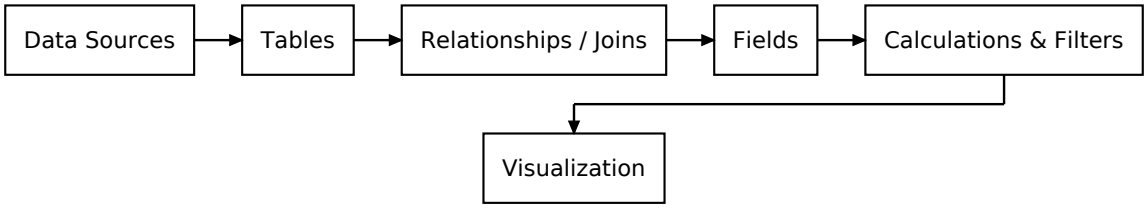


Figure: Data layer flow

Components of the data layer

1. Data sources (Excel, CSV, MySQL, SQL Server, Oracle, Google Sheets).
2. Tables (rows and columns).
3. Relationships (common fields such as Customer ID).
4. Joins (inner, left, right, full outer).
5. Unions (append similar tables).
6. Dimensions and measures.
7. Data types (text, number, date, Boolean).
8. Calculated fields (Profit Ratio = SUM([Profit]) / SUM([Sales])).
9. Filters (only 2026, one region).

Tables

- Structure: rows (records), columns (fields), field names, data values.
- Types: database tables, spreadsheet tables, logical tables (relationships), physical tables (joins/unions inside a logical table).

Combining tables

JOIN = adds COLUMNS (tables side by side, matched on ID)

Customer

ID	Name
101	Rahul
102	Priya

Sales

ID	Sales
101	50,000
102	30,000

→

Result (wider)

ID	Name	Sales
101	Rahul	50,000
102	Priya	30,000

UNION = adds ROWS (similar tables one below the other)

Jan Sales

Month	Sales
Jan	120

Feb Sales

Month	Sales
Feb	150

→

Result (longer)

Month	Sales
Jan	120
Feb	150

Figure: Join vs Union

- **Relationship**: logical link; tables stay separate; level of detail preserved.
- **Join**: adds columns from tables sharing a common field.
- **Union**: adds rows from tables with the same columns.

Data Source page operations

Select tables, create relationships/joins/unions, rename fields, change data types, filter, custom SQL, preview data.

Text tables

Tables can be shown as text tables when exact values are needed (Region, Product, Sales in rows and columns).

Conclusion

A well-built data layer with correct tables and links gives accurate dashboards.

Part E: Tableau Files

Q110. Explain Tableau workbooks and packaged files (.twb, .twbx, .tds, .tdsx, .hyper).

Introduction

Tableau saves work in different file types. A **workbook** holds the visual work; a **data source** file holds the connection; packaged versions add the data.

Tableau workbook

The main Tableau Desktop file used to create, save and share visualizations.

TABLEAU WORKBOOK (.twb / .twbx)

Data Sources	Worksheets	Dashboards	Stories
Excel, CSV, SQL...	one chart each	many sheets together	sequence of views

Also inside: Calculated fields · Filters · Parameters · Sets & Groups

.twbx additionally packs the data, images and other supporting files

Figure: What a Tableau workbook contains

Components: data sources, worksheets, dashboards, stories, calculated fields, filters, parameters, sets and groups.

Workflow:

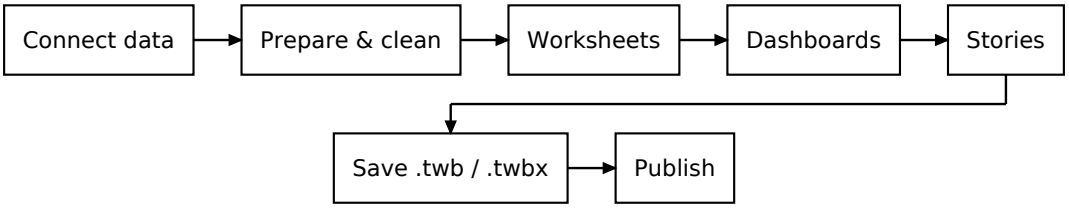


Figure: Typical workbook workflow

Packaged data source (.tdsx)

Contains the data source definition, a data extract and supporting files, so others can use it without access to the original database. Keeps calculated fields, groups, sets, hierarchies and metadata.

- Advantages: easy sharing, offline analysis, same data version for all, less dependency on external sources.
- Limitations: data outdated until refreshed, large file size, source changes not automatic.

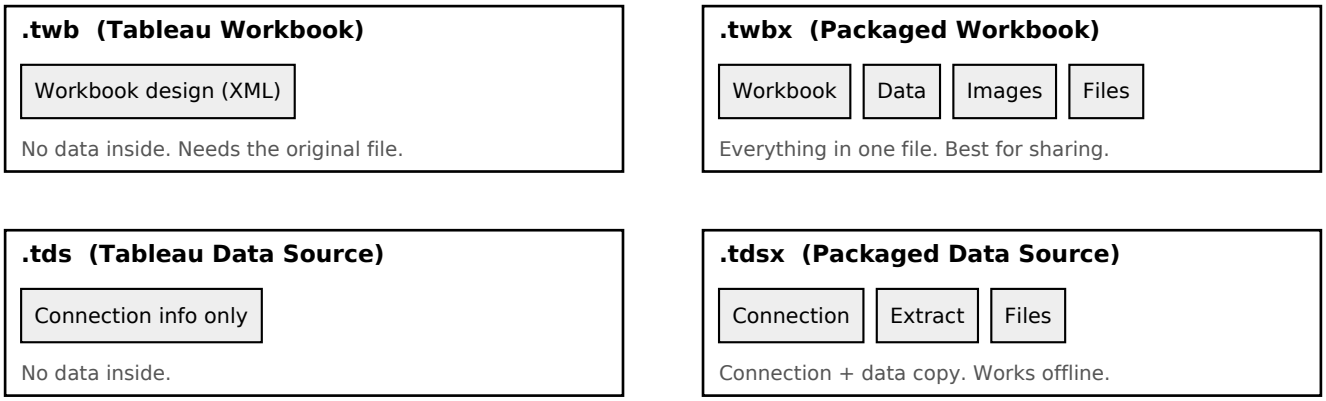


Figure: Plain files vs packaged files (the x means everything is packed inside)

Summary of file types

Extension	Name	Contains
.twb	Tableau Workbook	XML structure of sheets, dashboards, stories; no data
.twbx	Packaged Workbook	Workbook + data sources + images + supporting files
.tds	Tableau Data Source	Connection definition only
.tdsx	Packaged Data Source	Definition + extract + supporting files
.hyper	Extract file	The extracted data used by the Hyper engine

Conclusion

Use .twb/.tds when everyone has access to the data; use .twbx/.tdsx to share work together with its data.

Part F: The Tableau Workspace

Q111. Explain the Tableau workspace and its components with a neat diagram.

Introduction

The **Tableau workspace** is the main working environment where users connect to data, create visualizations, analyse information and build dashboards and stories.

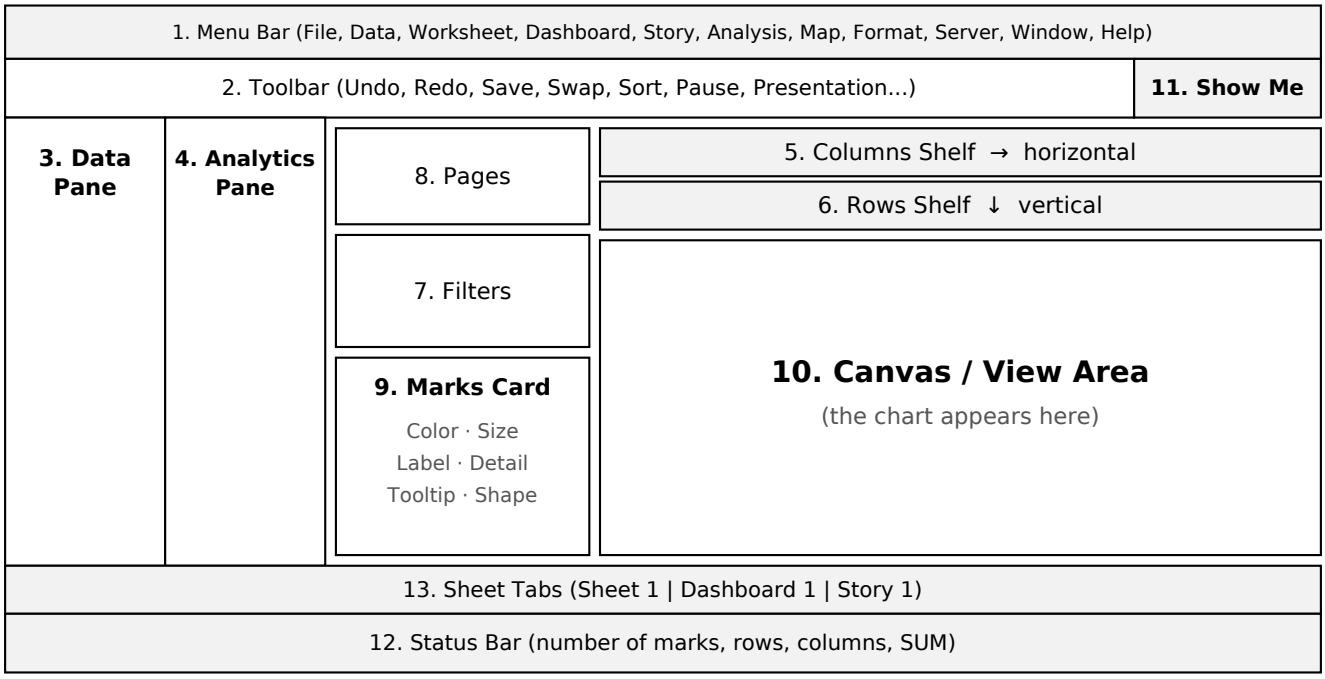


Figure: Layout of the Tableau workspace (numbers match the list)

Components

1. **Menu Bar**: commands for files, data, worksheets, dashboards, analysis, formatting and help (11 menus).
2. **Toolbar**: frequently used tools such as Undo, Redo, Save, Add Data Source, Sort, Swap, Presentation Mode.
3. **Data Pane**: dimensions, measures, parameters and sets of the connected source.
4. **Analytics Pane**: average line, median with quartiles, trend line, forecast, reference line, clusters.
5. **Columns shelf**: horizontal arrangement of fields.
6. **Rows shelf**: vertical arrangement of fields.
7. **Filters shelf**: restricts the data shown.
8. **Pages shelf**: divides the view into a sequence of pages.
9. **Marks Card**: Color, Size, Label, Detail, Shape, Tooltip.
10. **Canvas / View**: where the visualization is built and displayed.
11. **Show Me**: recommends chart types.
12. **Status Bar**: number of marks, selections and summary statistics.
13. **Sheet Tabs**: switch between worksheets, dashboards and stories.

Basic structure

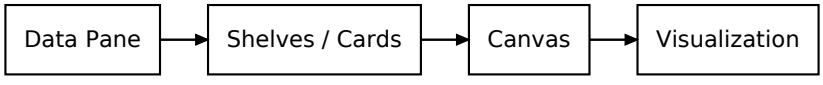


Figure: Basic structure of making a chart

Example: Sales to Columns and Category to Rows → Tableau automatically draws a bar chart.

Conclusion

The workspace is the complete environment for data analysis and for creating interactive visualizations, dashboards and stories.

Q112. Explain the Data Pane and its features in detail.

Introduction

The **Data Pane** is on the left side of the workspace. It displays connected data sources, tables, fields and metadata, and is the main area for accessing and organizing data before building charts.

Features

1. **Data Source**: shows connected sources; switch between them; live or extract.
2. **Tables**: lists tables; expand to see fields.
3. **Dimensions**: categorical fields (blue): Customer Name, Category, City, Region.
4. **Measures**: numeric fields (green): Sales, Profit, Quantity, Discount.
5. **Hierarchies**: drill-down structure, e.g. Country → State → City.
6. **Sets**: custom subsets to compare with the rest; fixed or dynamic.
7. **Groups**: combine values, e.g. North India = Delhi, Punjab, Haryana, Himachal Pradesh.
8. **Parameters**: user variables (numeric, string, date, Boolean) controlling calculations, filters, reference lines.
9. **Calculated fields**: Profit Ratio = SUM([Profit]) / SUM([Sales]).
10. **Geographic roles**: Country, State, Postal Code, Latitude, Longitude for maps.
11. **Default properties**: number/date format, aggregation, colour, aliases, comments.
12. **Search box**: find fields quickly.
13. **Sort options**: alphabetical or data source order.
14. **Metadata**: data type, default aggregation, geographic role, field name, source.

Data types

Type	Icon	Example
String	Abc	Customer Name
Number	#	Sales, Quantity
Date / Date & Time	Calendar (+ clock)	Order Date
Boolean	T/F	Is Returned
Geographic	Globe	Country

Common operations

Rename, hide/unhide, change data type, duplicate, create calculated fields, parameters, sets, groups, hierarchies and bins, assign geographic roles, split string fields.

Dimensions vs Measures

Dimension	Measure
Categorical, describes	Numeric, quantifies
Grouping, filtering, headers	Aggregation, axes
Blue (discrete)	Green (continuous)

Conclusion

The Data Pane organizes all fields in one place and makes drag-and-drop analysis, interactivity and large datasets easy to handle.

Q113. Explain the Analytics Pane: its features, how to use it and the types of analysis it supports.

Introduction

The **Analytics Pane**, next to the Data Pane, provides built-in analytical tools. Users **drag analytical objects onto a view** instead of writing calculations.

Features

1. **Constant line**: fixed reference value (target ₹1,00,000, pass mark 40, budget ₹5,00,000).
2. **Reference line**: average, median, minimum, maximum, sum or custom value.
3. **Distribution band**: shaded range showing variation.
4. **Box plot**: min, Q1, median, Q3, max; shows outliers.
5. **Trend line**: linear, logarithmic, exponential, polynomial, power.
6. **Forecast**: predicts future sales, revenue, demand, inventory.
7. **Cluster**: groups similar points using machine learning.
8. **Totals**: subtotals and grand totals.
9. **Average line**: compares values with the mean.
10. **Median with quartiles**: central tendency and spread.

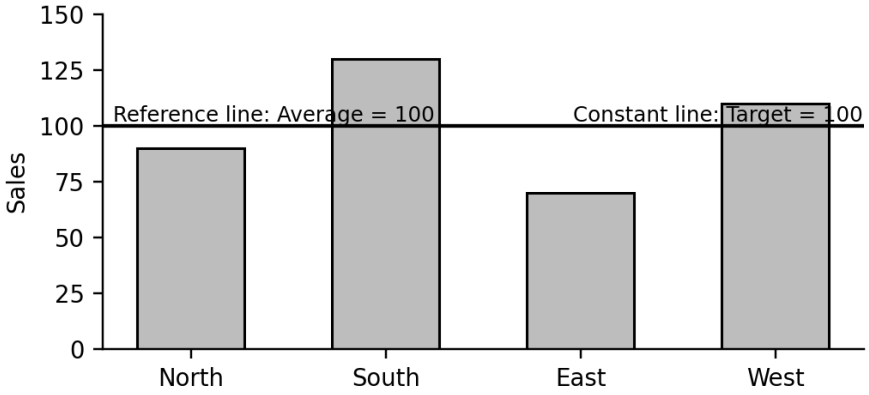


Figure: Constant line (fixed target) vs reference line (average of the data)

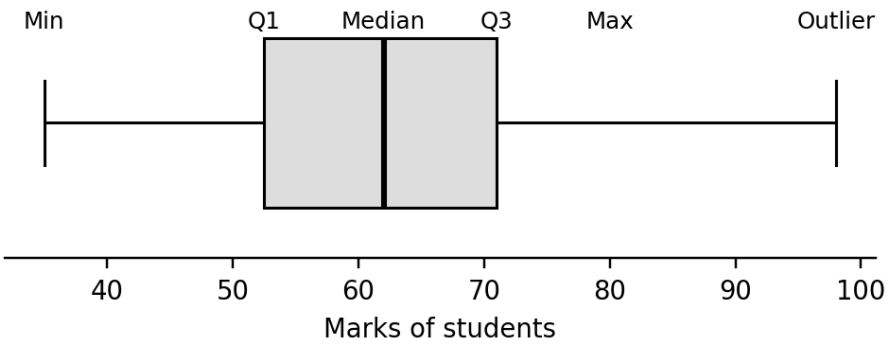


Figure: Box plot: Min, Q1, Median, Q3, Max and an outlier

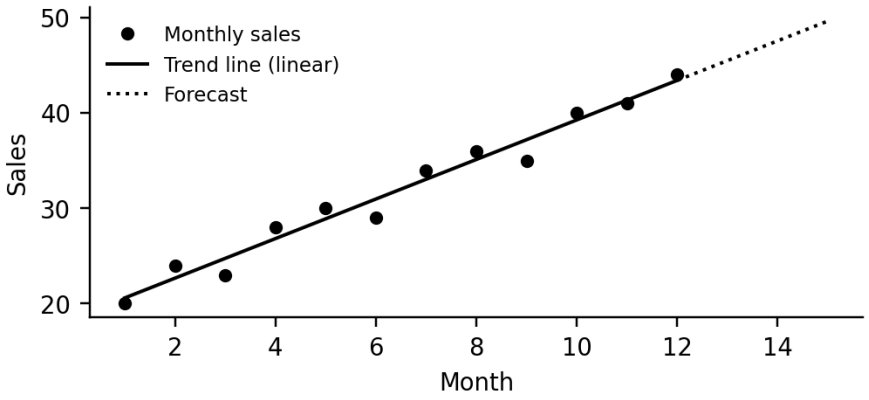


Figure: Trend line shows direction; forecast extends it into the future

How to use

1. Open a worksheet. 2. Click the Analytics tab. 3. Select an object. 4. Drag it onto the view. 5. Choose scope: **Entire Table, Pane or Cell**. 6. Configure formatting.

Types of analysis

Type	Tools
Statistical	Average, median, standard deviation, distribution, quartiles
Predictive	Forecasting, trend analysis, time-series analysis
Comparative	Reference lines, constant lines, distribution bands
Pattern recognition	Clustering, outlier detection, trend identification

Benefits and applications

No coding, statistical insight, forecasting, finds trends and outliers, easy benchmark comparison, faster decisions. Used in sales, finance, forecasting, healthcare, education, quality control, supply chain and marketing.

Conclusion

The Analytics Pane turns a simple chart into an analytical one with a few drag-and-drop actions.

Q114. Explain the Columns, Rows, Filters and Pages shelves in Tableau with examples.

Introduction

Shelves are areas of the workspace where fields are dropped to build and control a visualization.

1. Columns shelf

- Controls what appears **horizontally** (left to right).
- Dimension → column headers; measure → horizontal axis.

2. Rows shelf

- Controls what appears **vertically** (top to bottom).
- Dimension → row headers; measure → vertical axis.

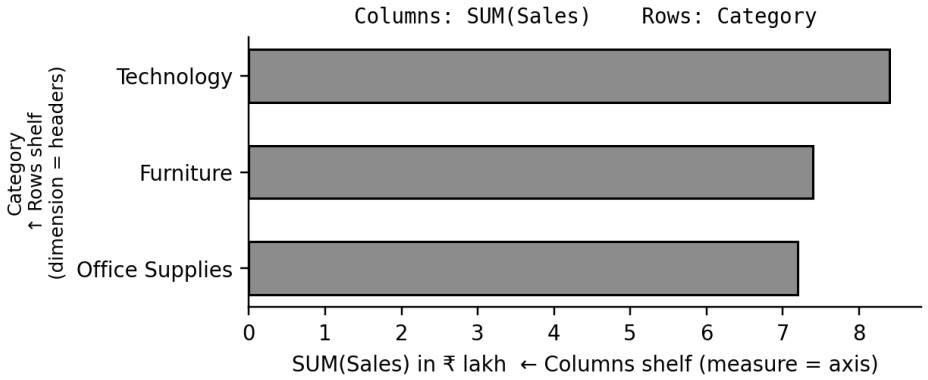


Figure: Sales on Columns + Category on Rows = horizontal bar chart

Rows	Columns	Result
Category	Sales	Bar chart of Sales by Category
Region	Sales	Sales by Region
Sales	Order Date	Sales trend over time

3. Filters shelf

- Restricts the data shown to what meets a condition.
- Types: dimension, measure, date, Top N, context and condition filters.
- Example: Country on Filters, select India → only Indian sales.

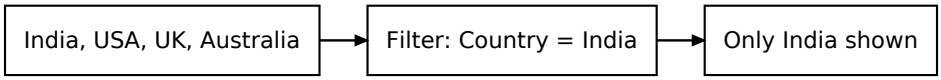


Figure: Filters shelf works like a sieve

4. Pages shelf

- Divides the view into sequential pages, one per value of a field.
- Play, Pause, Previous, Next controls give animation-like analysis.
- Example: Year on Pages → 2019 → 2020 → ... → 2024.

Differences

Columns	Rows
Horizontal	Vertical

Filters	Pages
Restricts data	Shows data page by page
Focused analysis	Sequential / time-based analysis
Filter card	Play / Next controls

Conclusion

Columns and Rows decide the **structure** of the chart; Filters decide **what data** is seen; Pages decide **how we move through** the data.